

2. **Hedge Funds:** long/short value-focused hedge fund (go long cheap stocks and short expensive stocks)
3. **Fixed Income:** 10-year Treasury bonds

How might such an investor choose to allocate across these three asset classes? Many advisors would recommend mean-variance optimization—the foundation for modern portfolio theory—which involves calculating returns of each asset, and correlations for each pair of assets, and solving for the efficient frontier that maximizes returns while minimizing risk. Remember, the inputs, based on history, translate into a forecast of returns and volatility for the future.

In the following tables, we highlight the fragile nature of complex models like mean-variance-optimization. In [Table 6.1](#), we calculate summary statistics for the S&P 500 total return index (stock market exposure, labeled SP500), the Fama and French long/short value factor (hedge fund exposure, labeled HML), and the 10-year Treasury bond (fixed income exposure, labeled LTR). We calculate monthly returns and related statistics over two subperiods: 1927 to 1960 and from 1961 to 2013. We choose two measurement periods to demonstrate how the choice of look-back period, which informs the input for the optimization algorithm, can drive dramatically different “optimized” allocations. In general, note that averages and standard deviations for these three asset classes don't differ dramatically, but the correlation matrix across the assets shows strong differences, despite being estimated over long time periods.